Adroit Technologies Protocol Driver

Name	Lovato RGK Series Serial
DLL	LOVRGK

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Contents

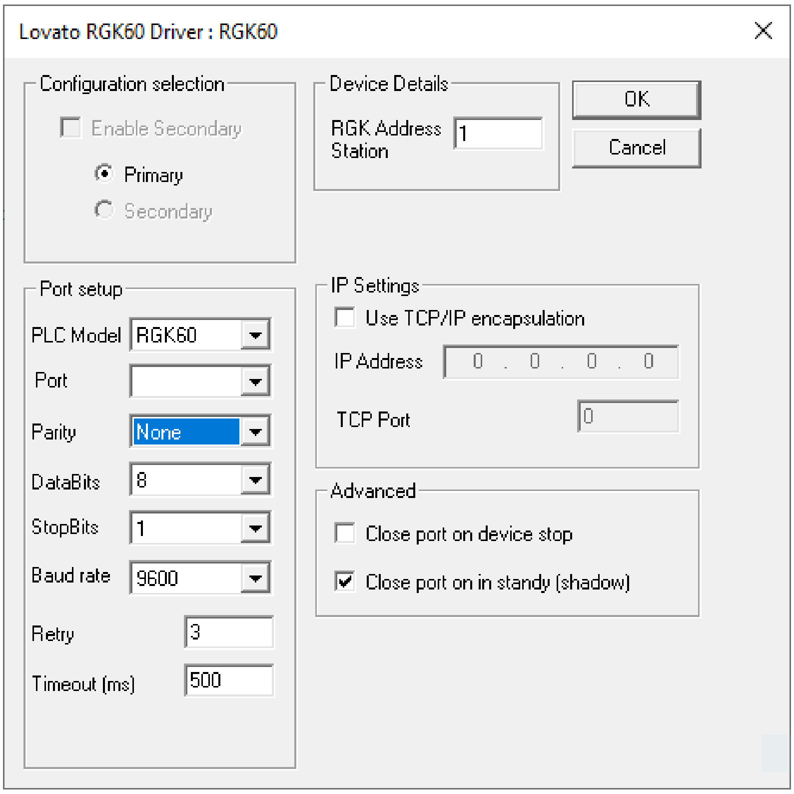
1. Overview

This document describes the communication protocol details and configuration implemented in the Adroit protocol driver, with the following logic units: Lovato RGK60, RGK800.

2. Wiring

Standard serial interconnection (RS-232-C).

3. Device Configuration



3.1. Configuration Selection (not available)

Enable Secondary – [Not implemented] Enables secondary configuration for dual redundant Controllers. This setting is disabled for this driver.

Primary / Secondary – [Not implemented] Select the primary or secondary button depending on which connection you wish to configure.

3.2. Device Details

RGK60 Station Address – Supply the unit address of the RGK60 device to communicate with.

3.3. Port Setup

PLC Model – Select the PLC model (RGK60 or RGK800).

Port – Select the serial port to use for communications.

Parity – Select even or uneven parity in a single serial transmission message.

DataBits – Select the number of data-bits per packet.

StopBits – Select the number of stop-bits per packet.

Baud Rate – The transmission rate of data (including start and stop bits). The default baud rate for an RGK60 unit is 9600.

Retry – The number of times to retry a transaction before a communication failure is reported.

Timeout – The number of milliseconds to wait between communication retries.

3.4. IP Settings

If an Ethernet gateway is used to provide an encapsulated Serial interface to the controller, the following settings are used to configure the connection:

Use TCP/IP encapsulation – Ticking this box switches the driver from communicating using a COM Port to using a TCP/IP Socket.

TCP Port – The TCP Port on which the driver should try to connect to the controller.

IP Address – The IP Address that will be used for communicating with the controller.

3.4.1. Advanced

Close port of device stop – If this check box is checked, when all logical devices sharing the same port have been stopped, the driver will release the COM port or IP port.

4. Supported Addresses

The Lovato RGK60 Adroit serial driver supports the following data address types:

Op-code	Description	Min Addr	Max Addr	Addressing example	Supported Adroit Datatypes
ST	General status request	1.1	1.12	ST 1.1 ST 1.12	Digital
ST	General status request	1	1	ST 1	Integer
NE	Error status request	1.1	4.5	ER 1.1 ER 4.5	Digital
NE	Error status request	1	4	ER 1 ER 4	Integer
MV	Mains voltage value (3-phase) request	1	6	MV 1 MV 6	Integer Real
GV	Genset voltage value (3-phase) request	1	6	GV 1 GV 6	Integer Real
SF	Mains & Genset frequency request	1	2	SF 1 SF 2	Real
LC	Load current (3-phase) request	1	3	LC 1 LC 3	Integer Real
MP	Mains total active power (3-phase) request	1	4	MP 1 MP 3	Real
MR	Mains total reactive power (3-phase) request	1	4	MR 1 MR 3	Real
MA	Mains total apparent power (3-phase) request	1	4	MA 1 MA 3	Real
MF	Mains power factor (3-phase) request	1	4	GF 1 GF 3	Real
GP	Genset total active power (3-phase) request	1	4	GP 1 GP 3	Real
GR	Genset reactive power (3-phase) request	1	4	GR 1 GR 3	Real
GA	Genset apparent power (3-phase) request	1	4	GA 1 GA 3	Real
GF	Genset power factor (3-phase) request	1	4	GF 1 GF 3	Real
EV	Mains & genset energy meter	1	4	EV 1 EV 4	Real
BV	Battery voltage request	1	1	BV 1	Real
HR	Engine total working time request	1	3	HR 1 HR 3	Integer Real
MN	Time-to-maintenance request (minutes)	1	1	MN 1	Integer Real
AV	Battery charger alternator voltage request	1	1	AV 1	Real
SV	Sensors value (pressure, temperature and fuel) request	1	4	SV 1 SV 3	Integer Real
WV	Engine speed request (RPM)	1	3	WV 1 WV 3	Integer
NS	Crank number request	1	2	NS 1 NS 2	Integer Real
PS	Front panel status	1.1	1.10	PS 1.1 PS 1.10	Digital
PS	Front panel status	1	1	PS 1	Integer
IS	Digital input / output status request	1.1	4.1	IS 1.1 IS 4.1	Digital
IS	Digital input / output status request	1	4	IS 1 IS 4	Integer
CW	Control Bits for simulation of the buttons on the controller	1.1	2.11	CW 1.1 CW 2.11	Digital

5. Protocol Definition and Considerations

5.1. Lovato RGK60 Message Structure

Message start	RGK Station Address	Op-code	Operator	Data	Start Checksum	Check-sum	Message Stop	Carriage-return linefeed
(	01	ST	?	DD…	&	CK	)	\r\n

Message start:
The opening bracket character (0x28) indicates the start of a new message.

RGK60 Station Address:
This two-digit number indicates the address of the RGK 60 device to which the message will to be sent.

Op-code:
The op-code identifies the type of information that is requested by the user. This could be any one of the op-codes described in the previous section ('Supported addresses').

Operator:
The operator indicates the type of message to send (Read or Write message). The Adroit Lovato RGK60 serial driver only supports 'Read' messages, thus the operator will always be a question mark ('?').

Data:
The data that the user wants to write to the unit. Because this driver does not support write functionality, the data portion will always be omitted.

Start checksum:
The ampersand character ('&') indicates the start of the message checksum.

Checksum:
The checksum is two characters wide and is the lowest significant byte of the algebraic sum of the single bytes that are transmitted, starting from the RGK60 station address to the last data byte.

Message stop:
The closing bracket (')') indicates the end of the message.

Carriage-return-linefeed:
The carriage-return and linefeed characters are appended to the end of each message.

5.2. General Status Request

The general status request message returns up to twelve bits of information, of which each of the bits indicates a certain named status within the unit. The entire block of bits can be requested by requesting the base address.

For example to request the entire address block, you would enter ST 1.
The different bits represent the following information:

Request	Status
ST 1.1	Bit 1 = TRUE if system is in OFF mode
ST 1.2	Bit 2 = TRUE if system is in MAN mode
ST 1.3	Bit 3 = TRUE if system is in AUT mode
ST 1.4	Bit 4 = TRUE if system is in TEST mode
ST 1.5	Bit 5 = TRUE if engine is running
ST 1.6	Bit 6 = TRUE if the alarm delay (oil+temp) has elapsed
ST 1.7	Bit 7 = TRUE if the mains voltage is present and within limits
ST 1.8	Bit 8 = TRUE if the generator voltage is present and within limits
ST 1.9	Bit 9 = TRUE if the mains contactor is closed
ST 1.10	Bit 10 = TRUE if the generator contactor is closed
ST 1.11	Bit 11 = TRUE if there are live alarms
ST 1.12	Bit 12 = TRUE if Automatic test is enabled

5.3. Error Status Request

Any error status ranging from 1.1 to 4.5 can be checked if it is ON by entering the 'NE' prefix and the error status number in the scan address field. For example, to return temperature warning status, you would enter NE 1.1. An entire address block can be requested by omitting the floating point and requesting the base address.

For example to request block 1, you would enter NE 1.

Request	Status
NE 1.1	Bit 1 = TRUE if engine temperature warning (analog sensor) alarm is active
NE 1.2	Bit 2 = TRUE if high engine temperature (analog sensor) alarm is active
NE 1.3	Bit 3 = TRUE if temperature analog sensor fault alarm is active
NE 1.4	Bit 4 = TRUE if high engine temperature (digital sensor) alarm is active
NE 1.5	Bit 5 = TRUE if oil pressure warning (analog sensor) alarm is active
NE 1.6	Bit 6 = TRUE if low oil pressure (analog sensor) alarm is active
NE 1.7	Bit 7 = TRUE if pressure analog sensor fault alarm is active
NE 1.8	Bit 8 = TRUE if low oil pressure (digital sensor) alarm is active
NE 1.9	Bit 9 = TRUE if pressure digital sensor fault alarm is active
NE 1.10	Bit 10 = TRUE if fuel level warning (analog sensor) alarm is active
NE 1.11	Bit 11 = TRUE if low fuel level (analog sensor) alarm is active
NE 1.12	Bit 12 = TRUE if level analog sensor fault alarm is active
NE 1.13	Bit 13 = TRUE if low fuel level (digital sensor) alarm is active
NE 1.14	Bit 14 = TRUE if high battery voltage alarm is active
NE 1.15	Bit 15 = TRUE if low batter voltage alarm is active
NE 1.16	Bit 16 = TRUE if inefficient battery alarm is active
NE 2.1	Bit 1 = TRUE if charger alternator failure alarm is active
NE 2.2	Bit 2 = TRUE if "W" signal failure alarm is active
NE 2.3	Bit 3 = TRUE if low engine "W" speed alarm is active

NE 2.4	Bit 4 = TRUE if high engine "W" speed alarm is active
NE 2.5	Bit 5 = TRUE if starting failure alarm is active
NE 2.6	Bit 6 = TRUE if emergency stop alarm is active
NE 2.7	Bit 7 = TRUE if unexpected stop alarm is active
NE 2.8	Bit 8 = TRUE if engine stop failure alarm is active
NE 2.9	Bit 9 = TRUE if low generator frequency alarm is active
NE 2.10	Bit 10 = TRUE if high generator frequency alarm is active
NE 2.11	Bit 11 = TRUE if low generator voltage alarm is active
NE 2.12	Bit 12 = TRUE if high generator voltage alarm is active
NE 2.13	Bit 13 = TRUE if generator asymmetry alarm is active
NE 2.14	Bit 14 = TRUE if generator short circuit alarm is active
NE 2.15	Bit 15 = TRUE if generator overload alarm is active
NE 2.16	Bit 16 = TRUE if external generator protection tripping alarm is active
NE 3.1	Bit 1 = TRUE if incorrect generator phase sequence alarm is active
NE 3.2	Bit 2 = TRUE if incorrect mains phase sequence alarm is active
NE 3.3	Bit 3 = TRUE if wrong system frequency setting alarm is active
NE 3.4	Bit 4 = TRUE if generator contactor failure alarm is active
NE 3.5	Bit 5 = TRUE if mains contactor failure alarm is active
NE 3.6	Bit 6 = TRUE if maintenance request alarm is active
NE 3.7	Bit 7 = TRUE if system error alarm is active
NE 3.8	Bit 8 = TRUE if fuel transfer empty alarm is active
NE 3.9	Bit 9 = TRUE if fuel transfer too full alarm is active
NE 3.10	Bit 10 = TRUE if rent hours exhausted alarm is active
NE 3.11	Bit 11 = TRUE if low radiator liquid level alarm is active
NE 3.12	Bit 12 = TRUE if circuit breaker closed alarm is active
NE 3.13	Bit 13 = TRUE if circuit breaker open alarm is active
NE 3.14	Bit 14 = TRUE if user alarm 1 is active
NE 3.15	Bit 15 = TRUE if user alarm 2 is active
NE 3.16	Bit 16 = TRUE if user alarm 3 is active
NE 4.1	Bit 1 = TRUE if user alarm 4 is active
NE 4.2	Bit 2 = TRUE if user alarm 5 is active
NE 4.3	Bit 3 = TRUE if user alarm 6 is active
NE 4.4	Bit 4 = TRUE if user alarm 7 is active
NE 4.5	Bit 5 = TRUE if user alarm 8 is active

5.4. Mains voltage (3-phase) Request

Each one of the 3 phases of voltages can be retrieved by entering 'MV' and the specific voltage you require.
For example, voltage across L1-L2 would be accessed by using MV 1.

Request	Return
MV 1	Voltage reading across L1-L2
MV 2	Voltage reading across L1-N
MV 3	Voltage reading across L2-L3
MV 4	Voltage reading across L2-N
MV 5	Voltage reading across L3-L1
MV 6	Voltage reading across L3-N

5.5. Generator Voltage Value Request

Each one of the 3 phases of voltages can be retrieved by entering 'GV' and the specific voltage you require.

Request	Return
GV 1	Voltage reading across L1-L2
GV 2	Voltage reading across L1-N
GV 3	Voltage reading across L2-L3
GV 4	Voltage reading across L2-N
GV 5	Voltage reading across L3-L1
GV 6	Voltage reading across L3-N

5.6. Mains and Generator Frequency Request

The mains or generator frequency value can be requested by using the address' SF 1/2.

Request	Return
SF 1	Mains frequency value.
SF 2	Generator frequency value.

5.7. Load Current Request

The load current expressed in Ampere, on L1, L2 and L3 can be requested by using the address' LC 1/2/3 respectively.

Request	Return
LC 1	Load current value on L1
LC 2	Load current value on L2
LC 3	Load current value on L3

5.8. Mains Active Power Request

The active power on the mains expressed in kW, for L1, L2 and L3 can be requested using the address' MP 1/2/3 respectively.

Request	Return
MP 1	Mains active power value on L1
MP 2	Mains active power value on L2
MP 3	Mains active power value on L3

5.9. Mains Reactive Power Request

The reactive power on the mains expressed in kVARkVAR for L1, L2 and L3 can be requested using the address' MR 1/2/3 respectively.

Request	Return
MR 1	Mains reactive power value on L1
MR 2	Mains reactive power value on L2
MR 3	Mains reactive power value on L3

5.10. Mains Apparent Power Request

The apparent power on the mains expressed in kVA for L1, L2 and L3 can be requested using the address' MA 1/2/3 respectively.

Request	Return
MA 1	Mains apparent power value on L1
MA 2	Mains apparent power value on L2
MA 3	Mains apparent power value on L3

5.11. Mains Power Factor Request

The mains power factor for L1, L2 and L3 can be requested using the address' MF 1/2/3 respectively.

Request	Return
MF 1	Mains power factor value on L1
MF 2	Mains power factor value on L2
MF 3	Mains power factor value on L3

5.12. Genset active Power Request

The active power on the genset expressed in kW for L1, L2 and L3 can be requested using the address' GP 1/2/3 respectively.

Request	Return
GP 1	Genset active power value on L1
GP 2	Genset active power value on L2
GP 3	Genset active power value on L3

5.13. Genset Reactive Power Request

The reactive power on the genset expressed in kVARkVAR for L1, L2 and L3 can be requested using the address' GR 1/2/3 respectively.

Request	Return
GR 1	Genset reactive power value on L1
GR 2	Genset reactive power value on L2
GR 3	Genset reactive power value on L3

5.14. Genset Apparent Power Request

The apparent power on the genset expressed in kVA, for L1, L2 and L3 can be requested using the address' GA 1/2/3 respectively.

Request	Return
GA 1	Genset apparent power value on L1
GA 2	Genset apparent power value on L2
GA 3	Genset apparent power value on L3

5.15. Genset Power Factor Request

The genset power factor for L1, L2 and L3 can be requested using the address' GF 1/2/3 respectively.

Request	Return
GF 1	Genset power factor value on L1
GF 2	Genset power factor value on L2
GF 3	Genset power factor value on L3

5.16. Energy Meter Request

The mains and genset active and reactive energy be requested using the EV address.

Request	Return
EV 1	Mains active energy meter (kW)
EV 2	Mains reactive energy meter (kVAR)
EV 3	Genset active energy meter (kW)
EV 4	Genset reactive energy meter (kVAR)

5.17. Battery Voltage Request

The battery voltage can be requested by using the address BV 1.

5.18. Engine Total Working Time (hours)

The engine working time is retrieved from the unit by using the HR address.

Request	Return
HR 1	Hours of engine working time
HR 2	Minutes of engine working time
HR 3 (Integer)	Hours and mins of engine working time represented in mins
HR 3 (Analog)	Hours and mins of engine working time represented in hours

5.19. Time-to-Maintenance Request (minutes)

The time-to-maintenance (in minutes) can be retrieved by using the address MN 1.

5.20. Battery Charger Alternator Voltage

The battery charger alternator voltage can be retrieved by using the address AV 1.

5.21. Sensor Value Request

The sensor value's for pressure, temperature and fuel can be requested using the address' SV 2/3/4 respectively.

Request	Return
SV 1	4 Bits. The first 3 bits indicate pressure, temperature and fuel level sensor availability respectively. The fourth bit indicates the unit of measurement to be in °C and bar.
SV 2	Pressure sensor value (bar)
SV 3	Temperature sensor value (˚C)
SV 4	Fuel level sensor value

5.22. Engine Speed (RPM) Request

The engine speed calculated using the genset frequency or using the W signal can be requested using the address' WV 1/2 respectively.

Request	Return
WV 1	Engine speed calculated using genset frequency
WV 2	Engine speed calculated using W signal
WV 3	unknown

5.23. Crank Number Request

The total number of crank (engine start) attempts and the percentage of successful cranks can be requested using the address' NS 1/2 respectively.

Request	Return
NS 1	The number of crank attempts
NS 2	The percentage of successful cranks.

5.24. Front Panel Status

The lit status (on/off) of the LED's on the front panel of the RGK60 device. When scanned as an integer the statuses are represented as individual bits packed into an integer type and must be marshalled to extract their statuses.

Request	Return
PS 1.1	Bit 1 = TRUE if RESET/OFF LED is lit
PS 1.2	Bit 2 = TRUE if MAN LED is lit.
PS 1.3	Bit 3 = TRUE if AUT LED is lit.
PS 1.4	Bit 4 = TRUE if TEST LED is lit.
PS 1.5	Bit 5 = TRUE if NET LED is lit.
PS 1.6	Bit 6 = TRUE if GEN LED is lit.
PS 1.7	Bit 7 = TRUE if ENGINEON LED is lit.
PS 1.8	Bit 8 = TRUE if mains contactor closed LED is lit.
PS 1.9	Bit 9 = TRUE if genset contactor closed LED is lit.
PS 1.10	Bit 10 = TRUE if Help LED is lit.

5.25. Digital Input / Output Status

The status of the I/O lines of the RGK60 can be retrieved by the following addresses:

Request	Status
IS 1.1	Bit 1 = TRUE if Terminal 8.1 is true
IS 1.2	Bit 2 = TRUE if Terminal 8.2 is true
IS 1.3	Bit 3 = TRUE if Terminal 8.3 is true
IS 1.4	Bit 4 = TRUE if Terminal 8.4 is true
IS 1.5	Bit 5 = TRUE if Terminal 8.5 is true
IS 1.6	Bit 6 = TRUE if Terminal 8.6 is true
IS 1.7	Bit 7 = TRUE if Terminal 8.7 is true
IS 1.8	Bit 8 = TRUE if Terminal 8.8 is true
IS 1.9	Bit 9 = TRUE if Terminal 8.9 is true
IS 1.10	Bit 10 = TRUE if Terminal 9.1 is true
IS 1.11	Bit 11 = TRUE if Terminal 9.2 is true
IS 1.12	Bit 12 = TRUE if Terminal 9.3 is true
IS 1.13	Bit 13 = TRUE if Terminal 4.1 is true
IS 1.14	Bit 14 = TRUE if Terminal 4.2 is true
IS 1.15	Bit 15 = TRUE if Terminal 5.3 is true
IS 1.16	Bit 16 = TRUE if Terminal 6.2 is true
IS 2.1	Bit 1 = TRUE if Terminal 6.3 is true
IS 2.2	Bit 2 = TRUE if Terminal 6.4 is true
IS 2.3	Bit 3 = TRUE if Terminal 6.5 is true
IS 2.4	Bit 4 = TRUE if Terminal 6.6 is true
IS 2.5	Bit 5 = TRUE if Terminal A.6 is true
IS 2.6	Bit 6 = TRUE If Terminal A.8 is true
IS 2.7	Unknown
IS 2.8	Bit 8 = TRUE if Expansion programmable input 1 is true
IS 2.9	Bit 9 = TRUE if Expansion programmable input 2 is true
IS 2.10	Bit 10 = TRUE if Expansion programmable input 3 is true
IS 2.11	Bit 11 = TRUE if Expansion programmable input 4 is true
IS 2.12	Bit 12 = TRUE if Expansion programmable input 5 is true
IS 2.13	Bit 13 = TRUE if Expansion programmable input 6 is true
IS 2.14	Bit 14 = TRUE if Expansion programmable input 7 is true
IS 2.15	Bit 15 = TRUE if Expansion programmable input 8 is true
IS 2.16	Bit 16 = TRUE if Expansion programmable input 9 is true
IS 3.1	Bit 1 = TRUE if Expansion programmable input 10 is true
IS 3.2	Bit 2 = TRUE if Expansion programmable input 11 is true
IS 3.3	Bit 3 = TRUE if Expansion programmable input 12 is true
IS 3.4	Bit 4 = TRUE if Expansion programmable input 13 is true
IS 3.5	Bit 5 = TRUE if Expansion programmable input 14 is true
IS 3.6	Bit 6 = TRUE if Expansion programmable output 1 is true
IS 3.7	Bit 7 = TRUE if Expansion programmable output 2 is true
IS 3.8	Bit 8 = TRUE if Expansion programmable output 3 is true
IS 3.9	Bit 9 = TRUE if Expansion programmable output 4 is true
IS 3.10	Bit 10 = TRUE if Expansion programmable output 5 is true
IS 3.11	Bit 11 = TRUE if Expansion programmable output 6 is true
IS 3.12	Bit 12 = TRUE if Expansion programmable output 7 is true
IS 3.13	Bit 13 = TRUE if Expansion programmable output 8 is true
IS 3.14	Bit 14 = TRUE if Expansion programmable output 9 is true

IS 3.15	Bit 15 = TRUE if Expansion programmable output 10 is true
IS 3.16	Bit 16 = TRUE if Expansion programmable output 11 is true
IS 4.1	Bit 1 = TRUE if Expansion programmable output 12 is true

5.26. Control Bits

Commands can be sent to the controller by simulating button presses on the front plate of the controller. Commands will only be sent on the rising edge of the digital control value. For this reason it is recommended that digital agents with pulse output configured is used for these scan addresses.

Request	Status
CW 1.1	Simulates pressing the "OFF" button on RGK60
CW 1.2	Simulates pressing the "MAN" button on RGK60
CW 1.3	Simulates pressing the "AUT" button on RGK60
CW 1.4	Simulates pressing the "TEST" button on RGK60
CW 1.5	Simulates pressing the "START" button on RGK60
CW 1.6	Simulates pressing the "STOP" button on RGK60
CW 1.7	Simulates pressing the "MAINS" button on RGK60
CW 1.8	Simulates pressing the "GEN" button on RGK60
CW 1.9	Simulates pressing the "HELP" button on RGK60
CW 1.10	Simulates pressing the "MINUS" button on RGK60
CW 1.11	Simulates pressing the "PLUS" button on RGK60
CW 1.12	Simulates pressing the "ARROW DOWN" button on RGK60
CW 1.13	Simulates pressing the "ARROW UP" button on RGK60
CW 1.14	Simulates pressing the "ENTER" button on RGK60
CW 1.15	Simulates pressing the "EXIT" button on RGK60
CW 1.16	Reset hardware
CW 2.1	Reset software
CW 2.2	Reset maintenance hours
CW 2.3	Reset run hours
CW 2.4	Simulates pressing the "Auto" button on RGK800
CW 2.5	Simulates pressing the "Manual" button on RGK800
CW 2.6	Simulates pressing the "Test" button on RGK800
CW 2.7	Simulates pressing the "Off" button on RGK800
CW 2.8	Simulates pressing the "Start" button on RGK800
CW 2.9	Simulates pressing the "Stop" button on RGK800
CW 2.10	Simulates pressing the "Gen" button on RGK800
CW 2.11	Simulates pressing the "Mains" button on RGK800

6. General Notes and Troubleshooting

6.1. Contact Details

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6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback (adroittechnologiesautomation.com)

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1	09-Jan-2022	K. Robinson	Updated device configuration descriptions, grammar changes	D. Jordaan
2.0				
3.0				

7.2. DLL Revision

Rev.	Date	Changes
	19-Dec-2012	First Release notes!
	23-Jan-2013	PS op-code was added
	13-Feb-2013	HR op-code updated
	06-Jun-2016	Added RGK800 simulation button controls CW 2.4 -> CW 2.11
21	31-Aug-2022	Additional stability fixes.